particle species with spin-½ there should be a partner species with spin-0; for every known particle species of spin-1, there should be a partner species with spin-½. For example, for the spin-½ electron there should be a spin-0 species called the *supersymmetric electron*, or *selectron* for short; for the spin-½ quarks there should be supersymmetric quarks, or squarks; for spin½ neutrinos there should be spin-0 sneutrinos; for spin-1 gluons, photons, and W and Z particles there should be spin-½ gluinos, photinos, and winos and *zinos*. (Yes, physicists get carried away.)

No one has ever detected any of these purported doppelgängers, and the explanation, physicists hope with fingers crossed, is that the supersymmetric particles are substantially heavier than their known counterparts. Theoretical considerations suggest that the supersymmetric particles could be a thousand times as massive as a proton, and in that case their failure to appear in experimental data wouldn't be mysterious: existing atom smashers don't have adequate power to produce them. In the coming decade this will change. Already, the newly upgraded accelerator at Fermilab has a shot at discovering some supersymmetric particles. And, as with the Higgs, should Fermilab fail to find evidence of supersymmetry and if the expected mass range of the supersymmetric particles is fairly accurate, the Large Hadron Collider should produce them with ease.

The confirmation of supersymmetry would be the most important development in elementary particle physics in more than two decades. It would establish the next step in our understanding beyond the successful standard model of particle physics and would provide circumstantial evidence that string theory is on the right track. But note that it wouldn't prove string theory itself. Even though supersymmetry was discovered in the course of developing string theory, physicists have long since realized that supersymmetry is a more general principle that can easily be incorporated in traditional point-particle approaches. Confirmation of supersymmetry would establish a vital element of the string framework and would guide much subsequent research, but it wouldn't be string theory's smoking gun.